Recorded condition burden and death status

Synthetic Delaware EHR analysis

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The Delaware synthetic dataset

1,753 synthetic records, with 253 recorded deaths (14.43%).

Synthea provides simulated health histories, not a representative sample of current Delaware residents.

The generator version and seed were not supplied.

Data and analysis workflow

Patient, condition, medication and encounter CSVs linked by patient ID.

107,376 condition rows, 191,649 medication rows and 173,900 encounter rows.

Validated joins and dates, distinct historical counts, Wilson intervals and exploratory logistic models.

Cohort and demographic definitions

875 female and 878 male records.

Fixed reference date: 4 October 2024, the latest observed encounter start.

Age at reference is approximately 30-75 years. For deceased records it is hypothetical attained age, not age at death.

Exclusions change the outcome distribution

Restricting to 8-40 descriptions retains 1,716 records and 231 deaths.

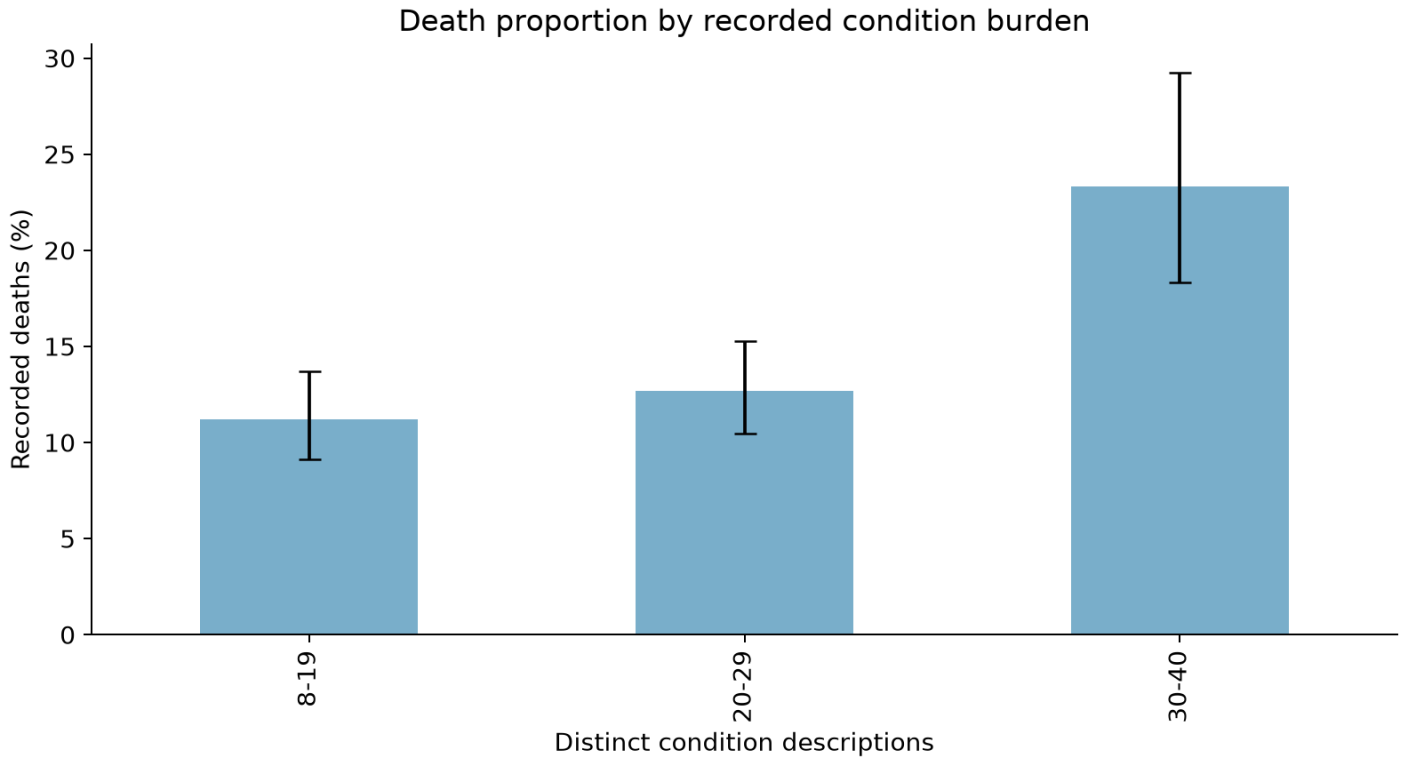
37 excluded records contain 22 deaths: 2.11% of records but 8.70% of deaths.

Death proportion falls by 0.97 percentage points. The full export remains the primary descriptive cohort.

Recorded burden and death proportion

Corrected groups: 8-19, 20-29 and 30-40 descriptions.

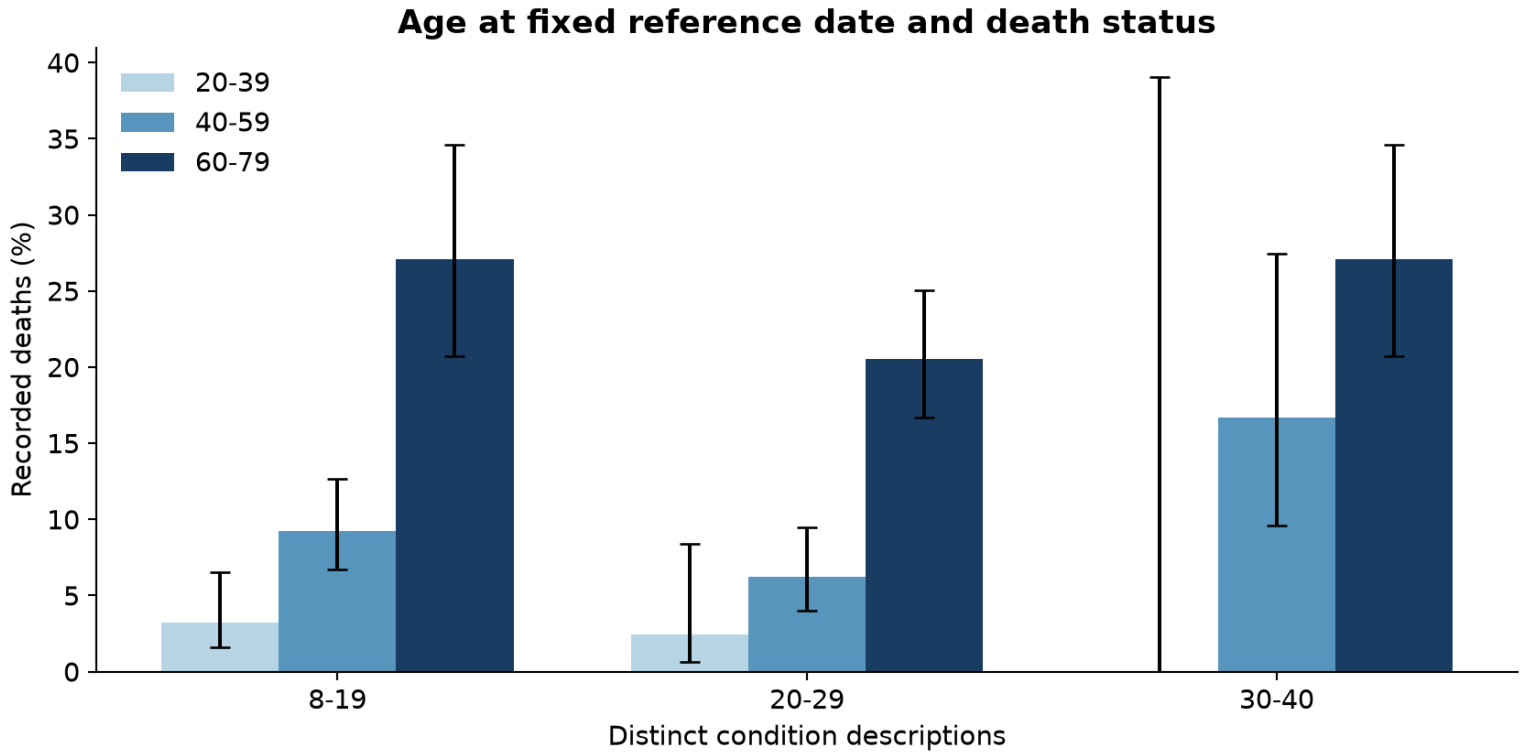
Restricted-cohort proportions: 11.22%, 12.68% and 23.35%.



Age at reference and death status

All nonempty groups show Wilson 95% intervals.

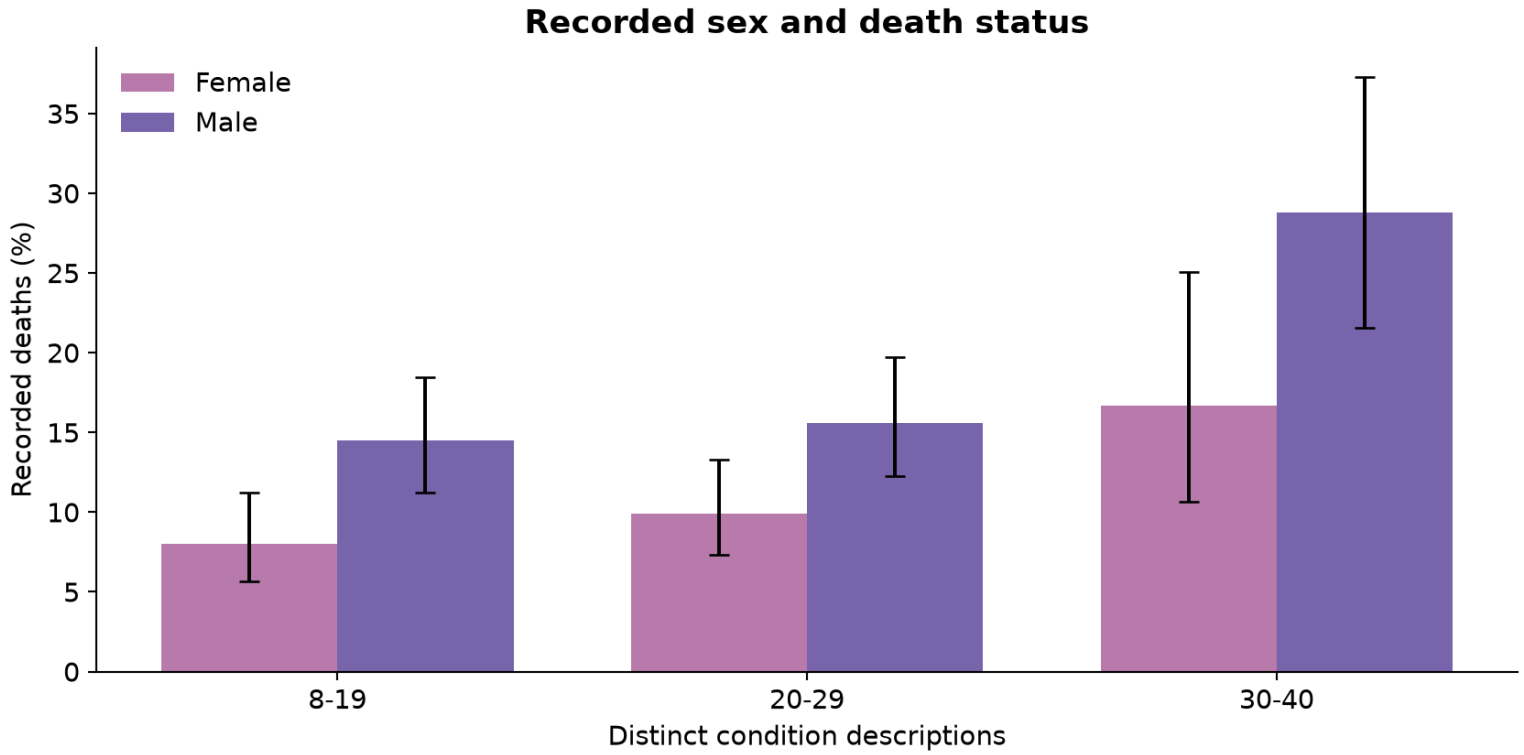
The youngest high-burden cell has 0/6 deaths, with an upper interval limit of 39.03%.



Recorded sex and death status

Male death proportions are higher within each restricted burden stratum.

Ratios: 1.81, 1.58 and 1.73. Corrected Holm p-values: 0.016, 0.037 and 0.037.



Condition counts and adjustment

Unadjusted condition OR: 1.019 in the full export and 1.039 in the restricted cohort.

Age/sex-adjusted condition OR: 0.967 in the full export (p = 0.002), compared with 0.983 (p = 0.159) in the restricted cohort.

Cohort definition and adjustment change the conclusion. Inverse coefficients do not demonstrate protection.

Sex-by-burden interaction

Restricted cohort: interaction OR 0.996 (95% CI 0.951-1.042), p = 0.847.

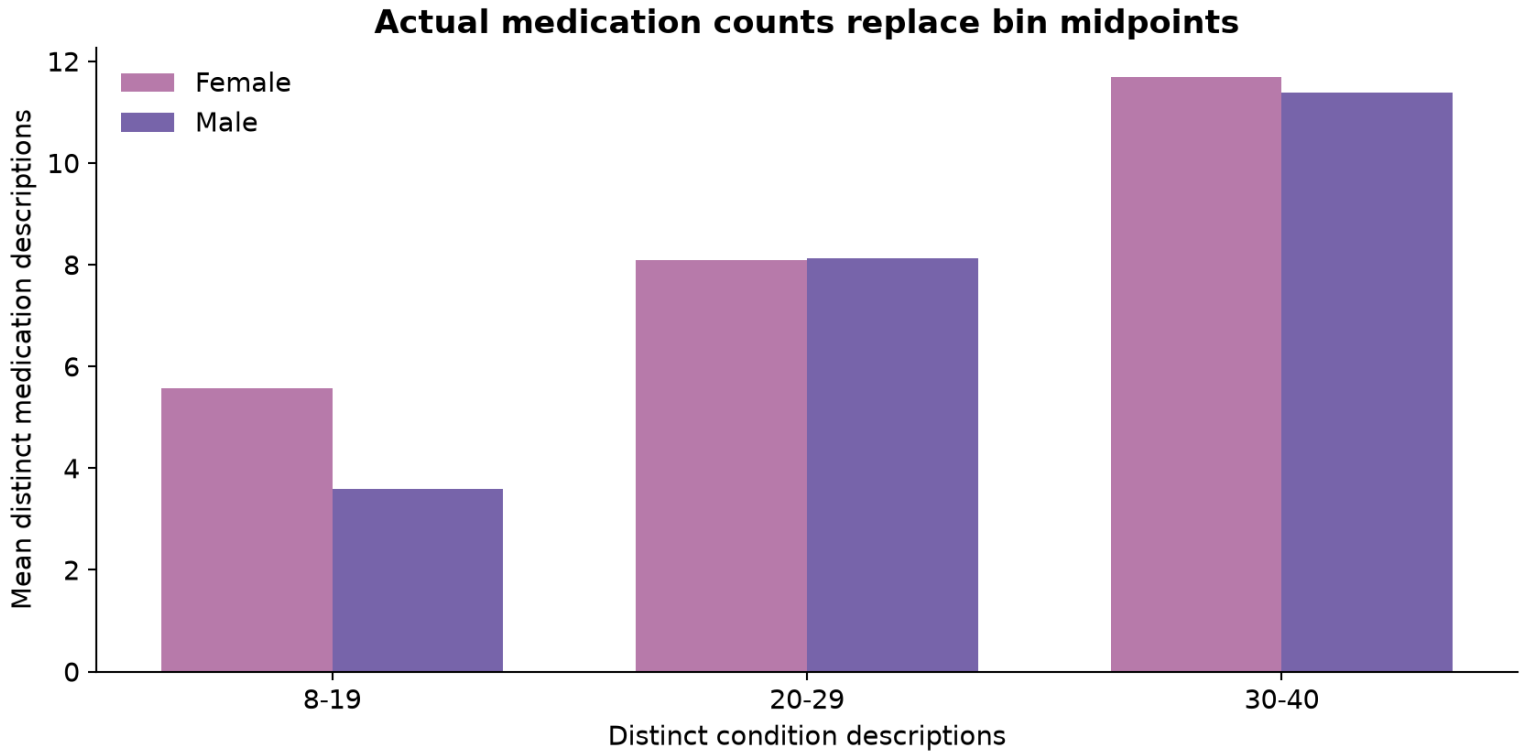
Full export: interaction OR 1.004 (0.964-1.045), p = 0.855.

Insufficient evidence of modification on the log-odds scale. This does not prove equal effects.

Medication history by recorded sex

Mean counts use actual distinct medication descriptions.

The lowest-burden means differ: 5.58 for females and 3.59 for males. Historical counts do not measure concurrent treatment.



Supported conclusions

Male-versus-female adjusted OR is about 1.84 in both cohorts.

Condition-count associations depend on cohort definition and adjustment.

The analysis does not establish whether severity, behaviour or physiological reserve explains these patterns.

Limits and further work

Synthetic data, unknown generation settings and historical exposures limit interpretation.

There are 25 deaths before age 20 despite adult reference ages. These are different age concepts.

A longitudinal study needs entry dates, baseline conditions, follow-up and validated diagnosis definitions.

References

Walonoski et al. (2018), JAMIA, 25(3), 230-238. doi:10.1093/jamia/ocx079

Fortin et al. (2012), Annals of Family Medicine, 10(2), 142-151. doi:10.1370/afm.1337

Crimmins, Shim and Zhang (2019), Clinical Chemistry, 65(1), 135-145. doi:10.1373/clinchem.2018.288332

Synthea CSV File Data Dictionary. Full links and provenance are in the essay and repository.

Appendix: methods and reproducibility

Included compressed extracts retain all rows and only required columns.

Run both notebooks, eight targeted tests and the report builder using Python 3.12.

Source/extract hashes, fixed cutoff and model formulas are documented. No clinical prediction claim is made.

Appendix: adjusted models

Full export / restricted cohort, respectively:

Male versus female: OR 1.836 / 1.835.

Age at reference, per year: OR 1.083 / 1.082.

Condition count, per description: OR 0.967 / 0.983.

Full confidence intervals, sample sizes and p-values are in outputs/models.csv.

Appendix: encounter composition

Acute-share Mann-Whitney comparisons use one Holm family across three burden strata.

Adjusted p-values: 1.000, 0.611 and 1.000. Routine share is complementary and is not tested twice.

The full export has 1,333 other-class encounters, excluded from the share denominator but included in total encounter counts.

Appendix: exploratory diagnosis flags

Among records matching at least one broad diagnosis category: females 70/656 deaths (10.67%), males 126/703 (17.92%).

Keyword categories overlap and do not measure severity. Benign or historical diagnoses may match broad terms.

Sparse race and disease cells require denominators and uncertainty, not causal explanations.

Appendix: medication history

Full-export death proportions are 9.68%, 8.97%, 10.72%, 16.95% and 19.69% across medication groups.

The pattern is not strictly monotonic. Association does not show medication effects on mortality.

